

Abstract

Acknowledgements

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Chapter 1

Strategic Vision and Project Scope

1.1 General Introduction

Digital transformation has significantly changed how educational institutions manage teaching, learning resources, assessment, and academic communication. However, many institutions still depend on fragmented combinations of messaging platforms, file-sharing tools, video-conferencing services, and manually coordinated assessment workflows. This fragmentation weakens the continuity of the learning experience, increases administrative burden, and makes it difficult to preserve a coherent academic record within one controlled environment.

Eduverse is an integrated educational platform developed to address this problem through a unified, role-aware, and multi-organization environment. Its main implemented client is a web application, supported by an MVP mobile companion application that extends access to core academic workflows for students and teachers. The platform treats the class as the central digital workspace in which communication, materials, assignments, examinations, results, notifications, live sessions, and academic support tools are managed in a coordinated manner.

From a technical perspective, the primary Eduverse client is implemented as a web application using Next.js, React, TypeScript, Tailwind CSS, and Supabase. Structured and frequently queried academic data are managed through Supabase PostgreSQL, while large files such as learning materials

and assignment submissions are handled through AWS S3. Real-time live sessions are supported through LiveKit, and AI-assisted interactions are integrated through OpenRouter. These technologies are introduced in this chapter only as a high-level overview; their analysis, design implications, and implementation details are deferred to later chapters.

Eduverse is intended for schools, universities, bootcamps, training centers, and other learning organizations that require coordinated digital management of classes, users, and academic activities. The platform supports multi-organization access and role-based responsibilities. Its main roles are organization owner, organization admin, teacher, and student. Within this context, Eduverse is positioned as a realistic educational platform that combines institutional control with day-to-day teaching and learning support.

1.2 Problem Statement

Educational institutions often rely on disconnected digital tools to carry out activities that are academically related but operationally separated. Materials may be shared through one service, class communication may take place in another, live teaching may depend on a third platform, and assessment records may be maintained elsewhere. As a result, students are required to move across multiple interfaces to complete what should be a single educational workflow, while teachers and administrators face difficulty maintaining visibility, consistency, and control.

The problem becomes more complex in institutions that operate multiple organizations, departments, or cohorts with different access requirements and instructional needs. A modern educational platform must not only support teaching activities, but must also manage user roles, organization boundaries, class membership, and feature availability in a controlled manner. Without such control, systems either become overly rigid or too loosely governed to support dependable academic administration.

Another important issue is the difference between primary and companion usage contexts. Rich authoring, administration, and system management workflows are best handled through a full web environment, yet students and teachers also need practical mobile access to high-frequency tasks such as checking classes, following notifications, viewing assignments, opening materials, joining live sessions, and participating in discussions. This creates a need for a platform that combines a comprehensive web client with a focused

mobile companion application rather than assuming that a single client type can serve all educational needs equally well.

1.3 Project Motivation

The development of Eduverse is motivated by the need to provide a unified academic environment in which core educational workflows can be carried out without fragmentation. A coherent class-centered platform can improve the experience of students, reduce administrative overhead for teachers and institutional staff, and support clearer tracking of academic progress, communication, and assessment outcomes. This motivation is both pedagogical and operational, since effective digital education depends not only on content delivery but also on well-organized workflow continuity.

The project is also motivated by the growing importance of flexible digital infrastructure in contemporary education. Institutions increasingly require platforms that can support multiple organizations, different user roles, and a wide range of learning activities within one extensible system. Eduverse responds to this need by combining class management, academic interaction, live teaching support, role-based access, and selected intelligent services within one platform while maintaining clear boundaries between core features and areas that remain under development.

From an academic and engineering perspective, Eduverse is valuable because it represents a realistic software engineering project that integrates frontend development, backend services, data management, real-time communication, and applied AI support in a single educational context. This makes the project suitable not only as a functional platform but also as a meaningful graduation project with clear analytical, technical, and documentation value.

1.4 Strategic Objectives

The strategic objectives of Eduverse are centered on establishing a reliable digital learning platform that supports institutional management and class-level academic workflows within a unified environment. These objectives define the project direction at both the system level and the user-experience level.

1.4.1 Project Aim

The main aim of Eduverse is to develop an integrated educational platform whose primary implemented client is a web application, complemented by an MVP mobile companion application, in order to centralize essential academic workflows for organizations, teachers, and students within a role-based and extensible system.

1.4.2 Project Objectives

The main project objectives are as follows:

- To centralize core educational workflows such as class management, materials distribution, assignments, examinations, results, notifications, communication, and live sessions within one coordinated platform.
- To support multi-organization operation and role-based access for organization owners, organization admins, teachers, and students.
- To provide a web-first operational environment that can support both institutional administration and day-to-day classroom activity.
- To extend selected high-frequency academic workflows to a mobile MVP companion application without claiming full feature parity with the web platform.
- To incorporate value-adding educational services such as an AI Agent and an integrated educational IDE at the platform level, while reserving deeper technical treatment for later chapters.
- To organize platform data and services in a manner that is maintainable, scalable, and suitable for academic institutions with different operational needs.
- To produce a system that is academically meaningful as a graduation project and practically relevant as a modern educational software platform.

1.5 Project Scope and Boundaries

The scope of Eduverse, at the current stage of this report, is defined around the implemented web platform and the existing mobile MVP companion application. Chapter 1 is limited to introducing the problem context, project vision, user groups, platform scope, and report organization. Detailed requirements models, architecture decisions, database design, implementation details, and testing evidence are intentionally reserved for the subsequent chapters.

1.5.1 Stakeholders and Role-Based Access

Eduverse serves several stakeholder groups within educational organizations. At the institutional level, the platform supports organization owners and organization admins who are responsible for configuring organizations, managing users, supervising access, and maintaining operational control across classes. At the instructional level, teachers manage course-related academic activities such as materials, assignments, live sessions, and evaluation. At the learner level, students participate in the classes to which they belong by accessing learning materials, tracking coursework, joining sessions, communicating with peers and instructors, and reviewing academic outcomes.

This role-based model is a central boundary of the system. Eduverse does not treat all users as functionally identical participants. Instead, access is governed according to organizational membership and academic responsibility. This approach supports clearer accountability, safer data exposure, and more appropriate separation between administrative, teaching, and learning tasks.

1.5.2 Functional Scope

Within its current implemented scope, Eduverse includes role-aware authentication, multi-organization access, class management, materials, assignments, examinations, results, notifications, live sessions, AI-assisted interaction, and an integrated educational IDE. These functions form the main operational identity of the platform and distinguish it from narrower systems that focus only on content delivery or communication.

At the data and service level, Eduverse also includes a deliberate separation between structured academic records and large file assets. Structured

and frequently queried data are handled through Supabase PostgreSQL, while larger files such as uploaded materials and submissions are stored through AWS S3. Live synchronous activity is supported through LiveKit, and AI requests are integrated through OpenRouter. In this chapter, these elements are mentioned only to define the platform scope at a high level rather than to explain their internal design.

The present chapter does not attempt to document database tables in detail, describe complete API behavior, or present the full system architecture. Those topics belong to later chapters and are outside the scope of the introductory discussion.

1.5.3 Web and Mobile Client Overview

The main implemented Eduverse client is the web application. It provides the fullest operational coverage of the platform, including multi-organization navigation, role-based access, class management, materials, assignments, examinations, results, notifications, live sessions, AI Agent workflows, and the integrated educational IDE. Because it is the primary working environment for both administration and rich instructional workflows, the web application should be regarded as the core client of the platform.

The mobile application is an MVP companion application built with Expo, React Native, and TypeScript. It is integrated with the same authentication and backend environment and currently supports practical academic workflows such as authentication, dashboard access, class viewing, assignments, notifications, chat, materials, and live sessions. However, the mobile client is not yet feature-complete relative to the wider Eduverse platform and should therefore be described as a companion application rather than a full replacement for the web system. Advanced mobile-side areas such as broader administration, feature-rich authoring, and full parity with the web platform remain under development.

1.5.4 Report Organization

The remainder of this dissertation is organized as follows. Chapter 2 presents the literature review and theoretical framework relevant to digital educational platforms and the technologies used by Eduverse. Chapter 3 describes the adopted development methodology and the general engineering process followed in the project. Chapter 4 analyzes the system domain and documents

the main requirements. Chapter 5 presents the system design, including the structural and interface-oriented design perspective. Chapter 6 discusses software and engineering implementation. Chapter 7 presents testing and evaluation. Chapter 8 is reserved for future work and planned extensions. The conclusion then summarizes the study and highlights the final academic outcomes of the project.

Chapter 2

Literature Review and Theoretical Framework

TODO: This chapter outline is reserved for a later documentation step.

- 2.1 Introduction**
- 2.2 Evolution of Learning Management Systems**
- 2.3 Limitations of Fragmented Digital Educational Environments**
- 2.4 Role-Based Digital Learning Platforms**
- 2.5 Online Examination Systems and Assessment Integrity**
- 2.6 Real-Time Collaboration and Live Learning**
- 2.7 Artificial Intelligence in Education**
- 2.8 Technical and Theoretical Foundations**
 - 2.8.1 Next.js and React**
 - 2.8.2 TypeScript**
 - 2.8.3 Supabase and PostgreSQL**
 - 2.8.4 LiveKit and Real-Time Communication**
 - 2.8.5 Role-Based Access Control**
- 2.9 Summary**

Chapter 3

Methodology

TODO: This chapter outline is reserved for a later documentation step.

3.1 Introduction

3.2 Adopted Development Methodology

3.2.1 Iterative / Agile Development

3.2.2 Justification of Technology Choices

3.3 Development Phases

3.3.1 System Planning and Requirements

3.3.2 UI/UX and Frontend Development

3.3.3 Backend and Database Development

3.3.4 Exams, Live Sessions, and Communication Features

3.3.5 Testing and Refinement

3.4 Modeling and Documentation Approach

3.5 Summary

Chapter 4

System Analysis and Requirements

TODO: This chapter outline is reserved for a later documentation step.

4.1 Introduction

4.2 Problem Domain Analysis

4.3 Stakeholder Needs Analysis

4.4 As-Is Analysis

4.5 To-Be Analysis

4.6 Requirements Engineering

4.6.1 Functional Requirements

4.6.2 Non-Functional Requirements

4.7 Use Case Modeling

4.7.1 Use Case Actors

4.7.2 Use Case Diagrams

4.7.3 Planned Extension Use Cases

4.8 Activity Diagrams

4.8.1 Access and Onboarding

4.8.2 Organization and Configuration Management

4.8.3 Classroom Operations

4.8.4 Assessment Workflows

4.8.5 Synchronous Learning and Follow-Up

Chapter 5

System Design

TODO: This chapter outline is reserved for a later documentation step.

- 5.1 Introduction
- 5.2 Architectural Design Overview
- 5.3 Data and Storage Design
- 5.4 Interface and Interaction Design
- 5.5 Integration and Security Design

Chapter 6

Software and Engineering Implementation

TODO: This chapter outline is reserved for a later documentation step.

6.1 Introduction

6.2 Web Application Implementation

6.3 Mobile Companion Implementation

6.4 Backend and Service Integration

6.5 Engineering Challenges and Resolutions

Chapter 7

System Testing and Evaluation

TODO: This chapter outline is reserved for a later documentation step.

7.1 Introduction

7.2 Testing Strategy

7.3 Functional Testing

7.4 Non-Functional Evaluation

7.5 Discussion of Results

Chapter 8

Future Work and Planned Features

TODO: This chapter outline is reserved for a later documentation step.

8.1 Introduction

8.2 Mobile Expansion

8.3 AI Agent and IDE Enhancement

8.4 Scalability and Institutional Growth

Chapter 9

Conclusion

TODO: Write the final conclusion after the main report chapters are completed.

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Appendix A

Appendices